

Markov Chains

1 Definitions

Definition 1.1 – Markov chain

A stochastic process (X_n) with values in I is called a Markov chain if for all $n \geq 0$ and $x_0, \dots, x_{n+1} \in I$, we have

$$\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, \dots, X_0 = x_0) = \mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n).$$

If $\mathbb{P}(X_{n+1} = y \mid X_n = x)$ is independent of n for all $x, y \in I$, then we call X a time-homogeneous Markov chain. Otherwise, it is called time-inhomogeneous.

In the case of time-homogeneous Markov chains, we can define

Definition 1.2 – Transition matrix

The transition matrix P is a square matrix of size $|I|$ with entries

$$P(x, y) = \mathbb{P}(X_1 = y \mid X_0 = x).$$

Definition 1.3 – Markov process

We say that the stochastic process (X_n) with values in I is Markov(λ, P) if it has initial distribution λ and transition matrix P . That is, if for all $n \geq 0$ and $x_0, \dots, x_{n+1} \in I$, we have

- (i) $\mathbb{P}(X_0 = x_0) = \lambda_{x_0}$
- (ii) $\mathbb{P}(X_{n+1} = x_{n+1} \mid X_n = x_n, \dots, X_0 = x_0) = P(x_n, x_{n+1})$

2 Basic properties

Theorem 2.1

The process X is Markov(λ, P) if and only if for all $n \geq 0$ and $x_0, \dots, x_n \in I$, we have

$$\mathbb{P}(X_0 = x_0, \dots, X_n = x_n) = \lambda_{x_0} p_{x_0 x_1} \dots p_{x_{n-1} x_n}.$$

Definition 2.1 – δ_i -mass

Let $i \in I$. The δ_i -mass at i is defined to be

$$\delta_{ij} = \mathbb{1}(i = j) = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise.} \end{cases}$$

Definition 2.2 – Independence

Let $X_1, \dots, X_n$ be n discrete random variables taking values in I . They are called independent if for all $x_1, \dots, x_n \in I$, we have

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) = \prod_{i=1}^n \mathbb{P}(X_i = x_i).$$

Let (X_n) be a sequence of discrete random variables with values in I . They are called independent if for all $k \geq 0$ and $i_1, \dots, i_k$ distinct and $x_1, \dots, x_k \in I$, we have

$$\mathbb{P}(X_{i_1} = x_1, \dots, X_{i_k} = x_k) = \prod_{l=1}^k \mathbb{P}(X_{i_l} = x_l).$$

Let (X_n) and (Y_n) be two sequences of discrete random variables taking values in I . They are called independent if for all $k, m \geq 0$ and $i_1, \dots, i_k$ distinct and $j_1, \dots, j_m$ distinct and $x_1, \dots, x_k, y_1, \dots, y_m \in I$, we have

$$\mathbb{P}(X_{i_1} = x_1, \dots, X_{i_k} = x_k, Y_{j_1} = y_1, \dots, Y_{j_m} = y_m) = \mathbb{P}(X_{i_1} = x_1, \dots, X_{i_k} = x_k) \mathbb{P}(Y_{j_1} = y_1, \dots, Y_{j_m} = y_m).$$

Theorem 2.2 – Simple Markov property

Suppose that X is Markov(λ, P) with values in I and let $m \in \mathbb{N}$. Then, conditional on $X_m = i$,

$$(X_{m+n})_{n \geq 0} \sim \text{Markov}(\delta_i, P)$$

and is independent of $X_0, \dots, X_m$.

Theorem 2.3

Suppose that X is Markov(λ, P). Then, for all $n, m \geq 0$ and $x, y \in I$, we have

- (i) $\mathbb{P}(X_n = x) = (\lambda P^n)_x$
- (ii) $\mathbb{P}(X_{n+m} = y \mid X_m = x) = p_{xy}(n)$ where $p_{xy}(n) = (P^n)_{xy}$

If all the eigenvalues of the transition matrix P are distinct, then

$$p_{xy}(n) = a_1 \lambda_1^n + \dots + a_k \lambda_k^n.$$

If a certain eigenvalue λ_i has algebraic multiplicity m , replace $a_i \lambda_i^n$ with $(b_0 + b_1 n + \dots + b_{m-1} n^{m-1}) \lambda_i^n$.

3 Communicating classes

Write $\mathbb{P}_x(\cdot)$ to denote $\mathbb{P}(\cdot \mid X_0 = x)$. Similarly, write $\mathbb{E}_x[\cdot]$ to denote $\mathbb{E}[\cdot \mid X_0 = x]$.

Definition 3.1 – Communicating states

Let X be a Markov chain with transition matrix P and values in I . For two states x and y in I , we say that x leads to y and denote it by $x \rightarrow y$ if

$$\mathbb{P}_x(X_n = y \text{ for some } n \geq 0) > 0.$$

We say that x communicates with y and denote it by $x \leftrightarrow y$ if $x \rightarrow y$ and $y \rightarrow x$.

Theorem 3.1

Let x and y be two states. The following are equivalent.

- (i) $x \rightarrow y$.
- (ii) There exists a sequence of states $x, i_1, i_2, \dots, i_k, y$ such that $p_{xi_1} p_{i_1 i_2} \dots p_{i_k y} > 0$.
- (iii) There exists $n \geq 0$ such that $p_{xy}(n) > 0$.

Corollary 3.1 – Communicating classes

The relation $\leftrightarrow$ defines an equivalence relation on I . The equivalence classes induced by $\leftrightarrow$ on I are called communicating classes.

Definition 3.2 – Closed communicating class

We say that a communicating class C is closed if whenever $x \in C$ and $x \rightarrow y$, we have $y \in C$.

Definition 3.3 – Irreducible transition matrix

A transition matrix P is called irreducible if it has a single communicating class. That is, for all $x, y \in I$, we have $x \leftrightarrow y$.

Definition 3.4 – Absorbing state

A state x is called absorbing if $\{x\}$ is a closed communicating class.

4 Hitting times and stopping times

Suppose that X is Markov chain with transition matrix P and with values in I . Let $A \subseteq I$.

Definition 4.1 – First hitting time

The first hitting time T_A is a random variable $T_A : \Omega \rightarrow \{0, 1, \dots\} \cup \{\infty\}$ given by

$$T_A(\omega) = \inf\{n \geq 0 : X_n(\omega) \in A\}$$

where the infimum of the empty set is taken to be ∞ .

Definition 4.2 – Hitting probability

The hitting probability of A is the function $h^A : I \rightarrow [0, 1]$ given by

$$h_i^A = \mathbb{P}_i(T_A < \infty).$$

Definition 4.3 – Mean hitting time

The mean hitting time of A is the function $k^A : I \rightarrow \mathbb{R} \cup \{\infty\}$ given by

$$k_i^A = \mathbb{E}_i[T_A] = \sum_{n=0}^{\infty} n \mathbb{P}_i(T_A = n) + \infty \cdot \mathbb{P}_i(T_A = \infty)$$

where $\infty \cdot 0$ is taken to be 0.

Theorem 4.1

The vector of hitting probabilities $(h_i^A : i \in I)$ solves the following system of linear equations.

$$h_i^A = \begin{cases} 1 & \text{if } i \in A \\ \sum_{j \in I} P(i, j) h_j^A & \text{otherwise.} \end{cases}$$

Moreover, (h_i^A) is the minimal nonnegative solution. That is, $0 \leq h_i^A \leq x_i$ for all $i \in I$ and nonnegative solutions (x_i) .

Consider a birth and death chain on $\mathbb{N}$ with transition probabilities

$$P(i, i+1) = p_i \text{ and } P(i, i-1) = q_i = 1 - p_i$$

for $i \geq 1$ and $P(0, 0) = 1$. We have that

$$h_i^{\{0\}} = \frac{\sum_{j=i}^{\infty} \gamma_j}{\sum_{j=0}^{\infty} \gamma_j} \text{ where } \gamma_j = \prod_{k=1}^j \frac{q_k}{p_k}.$$

Theorem 4.2

The vector of mean hitting times (k_i^A) is the minimal nonnegative solution to the linear system

$$k_i^A = \begin{cases} 0 & \text{if } i \in A \\ 1 + \sum_{j \notin A} P(i, j) k_j^A & \text{otherwise.} \end{cases}$$

Definition 4.4 – Stopping time

A stopping time T is a random variable $T : \Omega \rightarrow \{0, 1, \dots\} \cup \{\infty\}$ with the property that for every $n \in \mathbb{N}$, the event $\{T = n\}$ only depends on $X_0, \dots, X_n$.

Theorem 4.3 – Strong Markov property

Let X be $\text{Markov}(\lambda, P)$ and T be a stopping time. Then, conditional on $T < \infty$ and $X_T = i$,

$$(X_{T+n})_{n \geq 0} \sim \text{Markov}(\delta_i, P)$$

and is independent of $X_0, \dots, X_T$.

5 Transience and recurrence

Definition 5.1 – Transient and recurrent states and chains

A state $i \in I$ is called transient if

$$\mathbb{P}_i(X_n = i \text{ for infinitely many } n) = 0.$$

A state $i \in I$ is called recurrent if

$$\mathbb{P}_i(X_n = i \text{ for infinitely many } n) = 1.$$

A transition matrix P on I is called transient if all states $i \in I$ are transient. P is called recurrent if all states are recurrent.

Definition 5.2 – k -th return time

We define the successive times at which X visits i inductively. Set $T_i^{(0)} = 0$. For $k \geq 1$, set

$$T_i^{(k+1)} = \inf\{n \geq T_i^{(k)} + 1 : X_n = i\}.$$

Definition 5.3 – Total number of visits

The total number of visits to i is

$$V_i = \sum_{n=0}^{\infty} \mathbb{1}(X_n = i).$$

Lemma 5.1

For every $r \in \mathbb{N}$,

$$\mathbb{P}_i(V_i > r) = \mathbb{P}_i(T_i < \infty)^r.$$

Theorem 5.1

Let X be a Markov chain and $i \in I$. We have the following dichotomy.

- (i) If $\mathbb{P}_i(T_i < \infty) < 1$, then i is transient and $\sum_{n=0}^{\infty} p_{ii}(n) < \infty$.
- (ii) If $\mathbb{P}_i(T_i < \infty) = 1$, then i is recurrent and $\sum_{n=0}^{\infty} p_{ii}(n) = \infty$.

Theorem 5.2

Let $x, y \in I$ be two states that communicate. Then, they are either both transient or both recurrent.

Corollary 5.1

Either all states in a communicating class are transient or they are all recurrent.

Theorem 5.3

If C is a recurrent communicating class, then it must be closed.

Theorem 5.4

A finite closed communicating class is recurrent.

Theorem 5.5

Let P be irreducible and recurrent. Then, for all $x, y \in I$, we have

$$\mathbb{P}_x(T_y < \infty) = 1.$$

6 Random walks on $\mathbb{Z}^d$

Definition 6.1 – Simple random walk on $\mathbb{Z}^d$

A simple random walk on $\mathbb{Z}^d$ is a Markov chain with transition probabilities

$$\mathbb{P}(x, x + e_i) = \mathbb{P}(x, x - e_i) = \frac{1}{2d}$$

for all $x \in \mathbb{Z}^d$, where (e_i) is the standard basis of $\mathbb{R}^d$.

Theorem 6.1

The simple random walk on $\mathbb{Z}^d$ is recurrent when $d \leq 2$ and transient when $d \geq 3$.

Note that the asymmetric random walk on $\mathbb{Z}$ or any higher dimension is transient.

7 Invariant distribution

Definition 7.1 – Invariant distribution

A probability distribution $\pi = (\pi_i : i \in I)$ is called an invariant, or equilibrium, or stationary distribution for the Markov chain with transition matrix P if

$$\pi = \pi P.$$

Theorem 7.1

Let X be a Markov chain with transition matrix P and invariant distribution π . If $X_0 \sim \pi$, then $X_n \sim \pi$ for all n .

Theorem 7.2

Let I be a finite state space and suppose that there exists $i \in I$ such that for all $j \in I$,

$$p_{ij}(n) \rightarrow \pi_j \text{ as } n \rightarrow \infty.$$

Then, $\pi = (\pi_j : j \in I)$ is an invariant distribution.

Definition 7.2 – Expected visits during excursion

Let $k \in I$. The expected number of visits to i during an excursion from k is

$$v_k(i) = \mathbb{E}_k \left[\sum_{n=0}^{T_k-1} \mathbb{1}(X_n = i) \right].$$

Theorem 7.3

Suppose that P is irreducible and recurrent. Then, $v_k = (v_k(i) : i \in I)$ is an invariant measure. That is,

$$v_k = v_k P$$

with $0 < v_k(i) < \infty$ for all i and $v_k(k) = 1$.

Theorem 7.4

Suppose that P is an irreducible transition matrix. Let λ be an invariant measure satisfying $\lambda_k = 1$. Then, $\lambda_i \geq v_k(i)$ for all i . If P is also recurrent, then $\lambda = v_k$.

Definition 7.3 – Positive and null recurrent

Let $i \in I$ be a recurrent state.

- (i) We call i positive recurrent if $\mathbb{E}_i[T_i] < \infty$.
- (ii) We call i null recurrent if $\mathbb{E}_i[T_i] = \infty$.

Theorem 7.5

Suppose that P is an irreducible transition matrix. Then, the following are equivalent.

- (i) Every state is positive recurrent.
- (ii) Some state is positive recurrent.
- (iii) P has an invariant distribution π .

If any of the above holds, then for every $i \in I$, we have

$$\pi_i = \frac{1}{\mathbb{E}_i[T_i]}.$$

Corollary 7.1

Suppose that P is an irreducible transition matrix and has invariant distribution π . Then, for all $x, y \in I$, we have

$$v_x(y) = \frac{\pi(y)}{\pi(x)}.$$

8 Time reversibility

Proposition 8.1

Let X be a Markov chain with transition matrix P , which is irreducible and has invariant distribution π . Fix $N \in \mathbb{N}$ and suppose that $X_0 \sim \pi$. Then, the process $(Y_n)_{0 \leq n \leq N}$ defined via $Y_n = X_{N-n}$ is a Markov chain with transition matrix $\hat{P}$ given by

$$\hat{P}(x, y) = \frac{\pi(y)}{\pi(x)} P(y, x)$$

for all $x, y \in I$. Moreover, $\hat{P}$ is irreducible and has invariant distribution π .

Definition 8.1 – Time reversible Markov chain

We say that a Markov chain X with transition matrix P and invariant distribution π is time reversible if $\hat{P} = P$. That is, X is time reversible if and only if for all $x, y \in I$, we have

$$\pi(x)P(x, y) = \pi(y)P(y, x).$$

These are called the detailed balance equations.

Lemma 8.1

Suppose that μ is a distribution satisfying

$$\mu(x)P(x, y) = \mu(y)P(y, x)$$

for all $x, y \in I$. Then, μ is an invariant distribution for P .

9 Convergence to equilibrium

Definition 9.1 – Period and aperiodicity

Let P be a transition matrix and i be a state. The period of i is

$$d_i = \gcd\{n \geq 1 : p_{ii}(n) > 0\}.$$

We say that i is aperiodic if $d_i = 1$. P is aperiodic if all states are aperiodic.

Lemma 9.1

Let P be a transition matrix and i be a state. Then, $d_i = 1$ if and only if $p_{ii}(n) > 0$ for all n large enough.

Lemma 9.2

Let P be an irreducible transition matrix and i be an aperiodic state. Then, all states are aperiodic.

Theorem 9.1

Suppose that P is irreducible and aperiodic and has an invariant distribution π . Let $X \sim \text{Markov}(\lambda, P)$ where λ is a distribution. Then, for all $y \in I$, we have

$$\lim_{n \rightarrow \infty} \mathbb{P}(X_n = y) = \pi(y).$$

In particular, for any $x, y \in I$, we have

$$\lim_{n \rightarrow \infty} p_{xy}(n) = \pi(y).$$

Theorem 9.2

Suppose that P is an irreducible, null-recurrent and aperiodic transition matrix. Then, for all $x, y \in I$, we have

$$p_{xy}(n) \rightarrow 0 \text{ as } n \rightarrow \infty.$$

Theorem 9.3 – Ergodic theorem

Let $X \sim \text{Markov}(\lambda, P)$ where P is an irreducible and positive recurrent transition matrix that has invariant distribution π . Then, for all $x \in I$, we have almost surely

$$\frac{1}{n} \sum_{i=0}^{n-1} \mathbb{1}(X_i = x) \rightarrow \pi(x) \text{ as } n \rightarrow \infty.$$